

Assignment - 02

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Section : 10

CSE423

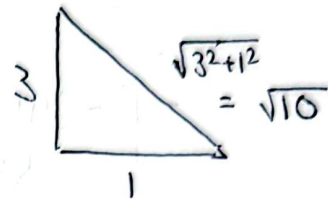
Ans. to the ques. no. 4

Here,

$$y = 3x - 8$$

$$m = 3, b = -8$$

$$\therefore \theta = \tan^{-1}(3) \\ = 71.56$$



$$\therefore \cos \theta = \frac{1}{\sqrt{10}} \quad \therefore \sin \theta = \frac{3}{\sqrt{10}}$$

Now,

$$M_R = T(0, b) R(\theta) \cdot M_x \cdot R(-\theta) \cdot T(0, -b)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -8 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\frac{1}{\sqrt{10}}) & -\sin(\frac{3}{\sqrt{10}}) & 0 \\ \sin(\frac{3}{\sqrt{10}}) & \cos(\frac{1}{\sqrt{10}}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\frac{4}{\sqrt{10}}) & \sin(\frac{3}{\sqrt{10}}) & 0 \\ -\sin(\frac{3}{\sqrt{10}}) & \cos(\frac{1}{\sqrt{10}}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.799 & 0.6001 & 0 \\ 0.6001 & 0.7998 & -8 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.799 & 0.6001 & 48 \\ 0.6001 & 0.7998 & -0.6 \\ 0 & 0 & 1 \end{bmatrix}$$

Now,

$$\cancel{P} = M \cancel{R}^T \times \cancel{P}$$

$$p = \begin{bmatrix} -0.799 & 0.6001 & 4.8 \\ 0.6001 & 0.7998 & -1.6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5.4035 \\ 4.1993 \\ 1 \end{bmatrix}$$

Ans. to the ques. no. 6

For shearing -

$$M_{shy} = \begin{bmatrix} 1 & 0 & 0 & -33 \\ 0 & 1 & 0 & 44 \\ 0 & 0 & 1 & -55 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 12 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 21 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 33 \\ 0 & 1 & 0 & -44 \\ 0 & 0 & 1 & 55 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 12 & 0 & -33 \\ 0 & 1 & 0 & 44 \\ 0 & 21 & 1 & -55 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 33 \\ 0 & 1 & 0 & -44 \\ 0 & 0 & 1 & 55 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 12 & 0 & -528 \\ 0 & 1 & 0 & 0 \\ 0 & 21 & 1 & -924 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Here,

$$P' = M_{shy} \times P$$

$$\therefore P = M_{shy}^{-1} \times P'$$

$$\therefore P = \begin{bmatrix} 1 & -12 & 0 & 528 \\ 0 & 1 & 0 & 0 \\ 0 & -21 & 1 & 924 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 125 \\ 255 \\ -674 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2407 \\ 255 \\ -5105 \\ 1 \end{bmatrix}$$

That is original coordinates -

$$P(-2407, 255, -5105)$$